HIMSCAN · SITE REPORT · UTTARAKHAND & HIMACHAL PRADESH

# Sample plot (Mussoorie ridge)

30.45980°N, 78.06440°E · Dehradun, Dehradun

Report built 2026-10-27 · data build checked 2026-10-27 17:01 · patch 883d102997fffff

## The answer

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| RELATIVE LANDSLIDE RANK | NATIONAL SUSCEPTIBILITY BAND | TERRAIN RULE: GROUND ≤25° | LIVABILITY |
|---|---|---|---|
| **Highest fifth**<br>elevated relative risk · percentile 98 | Severe<br>context only: about 65% of hill patches share the top band, so it does not separate one site from another | **27%**<br>of the patch; not shown to be safer | **74** /100<br>access, services, land, growth, setting, community; hazards are shown beside it, not subtracted |

**How to read the rank.** It comes from an independent slope-failure model tested on 135 landslides recorded after the model was frozen: AUC 0.70 (95% interval 0.66–0.74) against similar ground in the same district, 0.76 against road-matched ground. 79% fell in the highest-ranked fifth of the block's hill patches (42–50% for comparable ground). It is a *relative* screen, not a guarantee.

**Event-fitted relative risk: fourth fifth.** A model fitted to recorded landslides and floods, scored once on 256 events set aside before fitting: AUC 0.80 against similar ground (0.5 is chance; the gain over the physical score is modest, +0.08); 64% of those events fell in the highest-ranked fifth. It is a *relative* screen that also reflects where events get recorded (towns and roads), not a probability.

### Scores by driver

| DRIVER | PATCH SCORE (0–100) | WHAT IT COMBINES |
|---|---|---|
| Livability | **74** | access 96, services 86, land 23, growth 50, setting 96, community 100 |
| Landslide driver (physical) | **100** | susceptibility class share and steep ground; higher is worse |
| Flood driver (physical) | **0** | height above channels with a rainfall adjustment, past flooding, and the lake and landslide-dam outburst scenarios; higher is worse |
| Event-fitted relative risk | **Fourth fifth** (percentile 69) | landslide 95 and flood 50 (0–100 relative ranks, 100 = highest among all patches) fitted to recorded landslides and floods using terrain, channels, rainfall, soil, vegetation and built-up ground; the fifth compares this patch with similar ground in the same region, terrain, river and road setting. It includes where slope failures are recorded (towns and roads), so it is a ranking, not a probability |
| Earthquake driver | **12** | shaking expected once in 475 years at this ground type; it varies by region more than by site, so it is shown on its own |

Driver scores are fixed-weight combinations of measured layers. They rank patches against each other; they are not probabilities of an event.

### If the climate warms (scenario, not a forecast)

| DRIVER | TODAY | +1 °C | +2 °C | +3 °C |
|---|---|---|---|---|
| Flood driver (0–100) | **0** | 0 | 0 | 0 |

Assumptions: extreme rainfall rises 7% per degree (lower figure) or 15% per degree (upper figure, reported for mountain regions); river height scales with flow to the power 0.4, so the flood-prone band widens by the same factor; glacial lakes hold 25% more water per degree. Landslide susceptibility and earthquake shaking are not changed. These are stated assumptions, not predictions, and have no accuracy test.

## Hazard stack: what applies at your point

| LAYER | APPLIES TO | RESULT |
|---|---|---|
| Flood basin (river and torrent floors) | your point, 30 m | **above the flood basin** |
| Glacial-lake or glacier outburst corridor | your point, 30 m | **not on a traced corridor** |
| Observed monsoon inundation (satellite record) | patch | **upland patch, not scored** |
| Relative landslide rank | patch | **Highest fifth** |
| Seismic zone | region | **Zone VI, the highest category: follow the updated earthquake design code** |

The 30 m flood basin is the base layer for every plot, upland or plain; the other layers sit on top of it. Point rows describe your exact pixel, patch rows the whole ~0.8 km² hexagon.

**Among similar ground in this district** (same terrain class and elevation band): **Highest fifth** (percentile 94). Experimental: the district-level rank is still awaiting its own held-out confirmation.

Rule-based flags (context). Flags driven by the national landslide-susceptibility map did not separate event places from similar ground in our held-out tests, so they are not used for the rank.

| | |
|---|---|
| SEVERE | 100% of this patch sits in ILSM's high-or-worse landslide-susceptibility class (100% in the very-high class). |
| INFO | BIS's 2025 seismic map places the entire Himalayan arc in Zone VI (the highest category) -- construction here should follow the updated earthquake design code. |

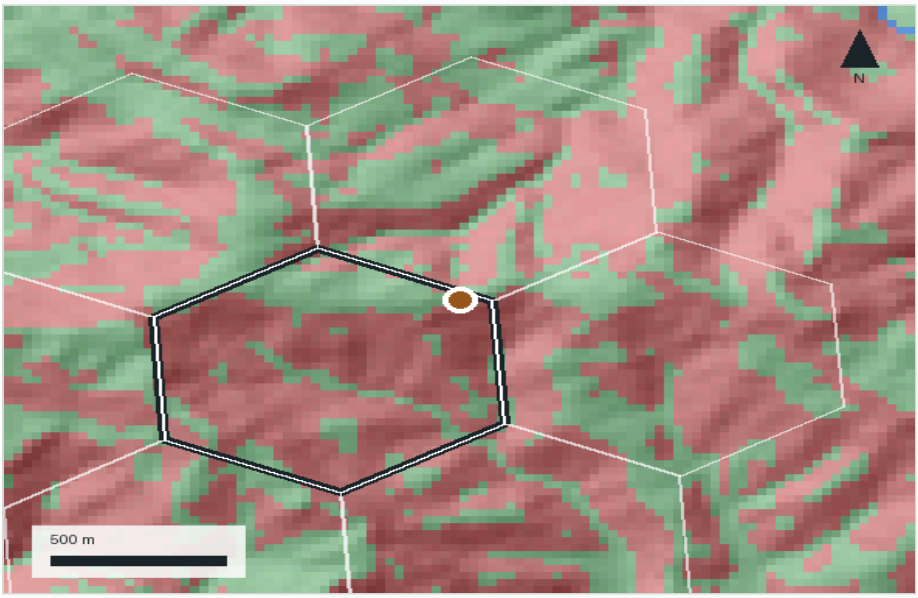
Surrounding patches: 6 Severe — worst nearby: Severe.

## Ground at 30 m

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### At your point (30 m, not averaged over the patch)

| | |
|---|---|
| Elevation | 2008 m |
| Position on the slope | ridge or spur |
| Slope at the point / within 150 m | 20.5° / 25.8° |
| Slope faces | south-east (141°) · towards the monsoon flow |
| Relief within 300 m | 234 m |
| 30 m flood basin at the point | **above the flood basin** |
| Height above nearest torrent channel | 504.7 m · no channel within 1 km |
| Height above nearest river | 1429.4 m · no river within 1 km |
| Upslope drainage area | 0.001 km² |




- Safe buildable ground
- Too steep (over 25°)
- Torrent or river floor
- Glacial-lake or glacier outburst corridor
- This patch (white inner line); surrounding patches outlined in white
- The point entered

2.6 km square window, 30.9 m pixels, over terrain shading. Classes use the same rules as the safe-buildable share above; the per-patch valley-confinement adjustment is not applied to individual pixels.

## Site and confidence

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| | |
|---|---|
| Coordinates entered | 30.45980, 78.06440 |
| Patch | 883d102997fffff · 0.785 km² |
| Point to patch centre | 487 m |
| Terrain | hill · 1,960 m · 28.8° |
| Measurements present | 14 of 14 |
| Tehsil | Dehradun · 855 patches |
| Safe ground, tehsil average | 69% |
| Legacy hazard bands in the tehsil | Severe 40% · High 17% · Moderate 13% · Low 31% |

A patch is an H3 hexagon of about 0.74 km². Everything below describes the whole patch containing this point; the map shows what varies inside it.

## Mountain hazards in depth

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| MEASURED ON THIS PATCH | VALUE | FLAG | RULE APPLIED |
|---|---|---|---|
| Safe buildable ground — 30 m pixels ≤25° and clear of water hazards | 27% | | Slope ≤25°, ≥5 m above torrents, ≥10 m above rivers, outside corridors |

| MEASURED ON THIS PATCH | VALUE | FLAG | RULE APPLIED |
|---|---|---|---|
| Ground steeper than 35° | 22% | no flag | Moderate when 50% of the patch |
| Within 30 m height of a river (100 km² catchment) | 0% | no flag | High at 10% within 30 m; Severe at 10% within 10 m |
| Within 5 m height of a torrent (2 km² catchment) | 0% | no flag | Moderate at 15%, High at 30% |
| Valley confinement (a gorge concentrates floods) | 1.01× | | 0.7× wide valley to 1.3× gorge; applied to river and torrent checks |
| On a traced glacial-lake outburst corridor (Hi-MAG lakes) | 0% | no flag | Flagged at 5%; Severe in the near half of the lake's reach |
| On a traced glacier debris / outburst corridor | 0% | no flag | High at 5% |
| Scenario reach: share of the patch flooded deeper than 0.5 m if an upstream glacial lake bursts<br>No modelled lake outburst path (mapped lakes of 5,000 m² and larger) reaches this patch within 40 km. Absence here is not evidence of safety: lakes formed after 2022, smaller lakes and landslide-dam outbursts are not covered | — | | Scenario: central value, low-high range in the note; share of 30 m cells in the patch that the flood covers deeper than 0.5 m |
| Recorded landslide, flood and related events within 10 km (2010-2026)<br>Within 5 / 10 / 25 km: 1 / 3 / 19. Nearest: 2011-08-28 landslide, 0.2 km (village); 2013-07-31 landslide, 8.7 km (village); 2014-08-16 landslide, 9.1 km (village); 2025-09-15 river flood, 10.9 km (place); 2022-08-20 flash flood, 12.4 km (village). Records come from news and official lists (2010-2026), located to village level at best; they favour roads and towns, so no record near a plot is not evidence that nothing happened. | 3 | | Count of recorded events within 10 km of the entered point; nearest five listed in the note |
| Landslide-dam breach discharge | 0.00 m³/s | no flag | High from 50 m³/s, Severe from 150 m³/s |
| Historical open water (JRC satellite record since 1984) | 0% | | Corroborates river flood when a share of the patch has been water |
| 100-year river flood depth, global model (WRI Aqueduct) | 0.00 m | | Context only — about 1 km resolution |
| Documented disaster incidents on this patch (severity score) | 0.0 | no flag | Moderate above 0.1, High from 2.0, Severe from 3.5 |
| Cloudburst hotspot rate (20 weather stations, Uttarakhand only) | 0.00 | | Context only — too few stations to describe an area |
| Legacy, context only: Landslide susceptibility — share of the patch high or very high (ILSM, IIT Delhi) | 100% | Severe | Moderate at 20%, High at 50%; Severe when 50% is very high |
| Legacy, context only: ...of which very high | 100% | | |

## Observed monsoon inundation and repeat-flood exposure (satellite record)

Upland patch (above about 900 m, not plains or foothill): not scored. River-valley flooding in the hills is covered by the torrent and river rows above.

## Vegetation and land change, 2014–2025 (satellite)

| MEASURE | THIS PATCH |
|---|---|
| Vegetated ground that lost greenness (a drop of 0.15 or more against its own earlier level) | **3%** |
| Of that ground, regrown by 2025 | **100%** |
| Lost greenness and not regrown (all years / 2022–2025) | **0% / 0%** |
| Greenness trend (NDVI per year) | **+0.004** |

Landsat 8 and 9, median of 15 August to 31 October for each year, 100 m, with cloud, snow and shadow masked; 90 vegetated pixels in this patch. A drop can come from a landslide scar, clearing, construction, fire or a dry year, and regrowth does not prove the ground has settled; this is a land-change signal, not a landslide map. Ground lost and not regrown in recent years raises the landslide driver slightly.

## Access and services

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| MEASURE | VALUE |
|---|---|
| Nearest motorable road | 0.0 km |
| Nearest highway | 2.4 km |
| District capital | 26.2 km |
| Major commercial centre | 26.2 km |
| Nearest hospital | 4.2 km |
| Nearest school | 3.6 km |
| Shops, banks, restaurants within 1 km | 9 places |
| ...within 5 km | 51 places |
| Population density | 2,159.4 people/km² |
| Building growth, 2016–2023 | 5.2 %/yr |
| Night-light growth, 2014–2025 | 1.33 × |

Distances in the table above follow the road network. Building growth is left blank where the 2023 building record has a gap.

### Road class, electricity and banks (straight-line, OpenStreetMap)

| MEASURE | VALUE |
|---|---|
| Nearest major road (national / state highway class) | 0.4 km |
| Nearest secondary road | 6.1 km |
| Nearest tertiary road | 0.3 km |
| Nearest local road or track | 0.0 km |
| Road length inside the patch: major / secondary / tertiary / local | 0.6 / 0.0 / 1.0 / 5.8 km |
| Nearest mapped power line (OSM, transmission and distribution) | 13.7 km |
| Nearest mapped substation (OSM) | 11.5 km |
| Nearest bank or ATM | 1.2 km |
| Banks and ATMs within 5 km | 10 |

| MEASURE | VALUE |
|---|---|
| Nearest railway line (OSM, straight-line) | 15.8 km |
| Nearest railway station or halt (OSM, straight-line) | 16.0 km |

Straight-line distances from the patch centre, not travel time. OpenStreetMap is patchy for minor roads and power lines in remote hills, so a missing feature may be unmapped, not absent. Whether a village is electrified, and the quality of supply, are not in open data and should be checked locally.

## Climate, weather and water

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33 measured rows for this patch, in the grammar global asset-risk reports use so the two can be read side by side. Each row shows the value in its own unit and a fixed 1–5 score (1 = low, 5 = very high); thresholds are fixed, so scores are comparable between sites. These are context for the site and for planning, not predictions of events.

| HAZARD | TYPE | VALUE | UNIT | SCORE 1–5 |
|---|---|---|---|---|
| CONTEXT | | | | |
| Worldcover 2020 | context | tree | class | – |
| World Settlement Footprint Evolution | context | 0.01 | share | – |
| SEISMIC | | | | |
| Earthquake | zone | VI | BIS zone | 5 |
| Peak Ground Acceleration | zone | 0.19 | g | 3 |
| Soil & Shaking | zone | 624.3 | m/s | 3 |
| Distance to Active Faults | zone | 21.86 | km | 3 |
| Liquefaction | zone | 0.0 | 0-1 | 1 |
| Quake Landslide | zone | 0.0 | cm | 1 |
| SOLID MASS | | | | |
| Avalanche | acute | 0.0 | share | 1 |
| Subsidence | acute | 26.7 | % | – |
| Soil Erosion | chronic | 36.72 | t/ha/yr | 4 |
| Soil Degradation | chronic | 2 | processes (0-5) | 3 |
| Solifluction | chronic | 66.3 | days/yr | 3 |
| TEMPERATURE | | | | |
| Heat Wave | acute | 1.67 | days/yr | 2 |
| Cold-Frost | acute | 66.3 | days/yr | 3 |
| Wildfire | acute | 0.0 | % of patch/yr | 1 |
| Fire Weather Stress Index | acute | 22.08 | FWI | 3 |
| Changing Temperature | chronic | 0.07 | C | – |
| Heat Stress | chronic | 29.46 | C | 1 |
| Heat-Humidity Stress Index | chronic | 18.09 | C | 1 |

| HAZARD | TYPE | VALUE | UNIT | SCORE 1–5 |
|---|---|---|---|---|
| Temperature Variability | chronic | 10.9 | C | 2 |
| Permafrost Thawing | chronic | 0.0 | % of area | 1 |
| WATER | | | | |
| Hail | acute | 0.8 | days/yr | 3 |
| Drought | acute | 10.28 | % of months | 4 |
| Heavy Precipitation | acute | 116.58 | mm | 3 |
| Flood | acute | 1 | 1-5 | 1 |
| River Flood (Defended) | acute | — | years | 1 |
| Changing Precipitation Patterns | chronic | 0.46 | % | – |
| Precipitation Variability | chronic | 0.82 | index | 4 |
| Water Stress | chronic | 4 | Aqueduct category | 5 |
| Water Scarcity | chronic | 3 | Aqueduct category | 4 |
| WIND | | | | |
| Storm | acute | 84.44 | km/h | 2 |
| Changing Wind Patterns | chronic | -2.65 | % | – |

### Long-run trends

| | |
|---|---|
| Warming trend (ERA5, full record) | 0.0167 °C/yr |
| Monsoon rainfall variability (ERA5-Land, coefficient of variation) | 0.221 |
| Vegetation health (MODIS NDVI, annual mean) | 0.544 |

A 10-year outlook (warming-trend based) is in development and is not part of this report.

## Can I buy, can I build

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| CHECK | STATUS | DETAIL |
|---|---|---|
| Seismic design | disclosed | BIS's 2025 seismic map places the whole Himalayan arc in Zone VI, the highest category — build to the updated earthquake code. |
| Who may buy this land | not assessed yet | Notified rules to be checked against the gazette; not part of this report. |
| River setbacks | not assessed yet | Distance rules from rivers; not part of this report. |
| Eco-sensitive zones | not assessed yet | Notified zone boundaries; not part of this report. |

This report is not a legal opinion. Ownership and title must be checked separately, one plot at a time.

## How good is this

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> The data build behind this report passed its automated checks on 2026-10-27 17:01 (PASS, build score 100, 0 failed checks).

**Event-fitted score.** The event-fitted relative risk was fitted on 431 recorded landslides and floods in Himachal Pradesh and Uttarakhand and tested once on 256 other events, set aside before fitting and scored once for the decision (events were split by map cell, so neighbouring events stay on one side; the working events were inspected over three rounds of changes). Against similar ground with no record that the model had not seen, the AUC is 0.80 (resampled by map cell: 0.76 to 0.86; 0.5 is chance); the physical score on the same events is 0.72, and the gain is +0.08 (interval +0.01 to +0.15 by map cell). One map cell holds a quarter of the events. 64% of the 256 events fell in the highest-ranked fifth (20% by chance) and 2% in the lowest fifth; 89% fell on the half of the land ranked highest and none in the lowest 1.7%. The gain holds for landslides and floods, before and after 2018, and for news and official records, and is smallest for the Geological Survey of India mapped slides in Himachal (older inventory slides). The test covers 10 map cells and the score uses built-up ground, so records near roads and towns raise it: read it as ranking evidence, not as probabilities. 252 of the 256 events were on hill or high-mountain terrain and 4 on plains, so the event-fitted score is shown for sloping hill land only.

**What was tested.** The landslide rank was trained on the Geological Survey of India landslide inventory (Himachal) and tested on 135 landslides recorded after the model was frozen (74 from news and other reports, 61 from Uttarakhand government records; none used in training). Across the block, 79% fell in the highest-ranked fifth of hill patches (42–50% for comparable ground). Against similar ground in the same district the AUC is 0.70 (95% interval 0.66–0.74, resampled by district and year; 0.5 is chance): 0.67 on the 74 news events, 0.74 on the 62 government records; one batch of 18 events scored lower (0.56). So the rank is strong at the regional level and moderate for telling one plot from its neighbours. Samples are small and news reports favour roads and towns, so read these as direction, not exact rates.

**Checked against prices.** In 300 localities with listings (residential plots from several portals; the largest portal is the guide and other listings are kept only within a factor of four of it; prices relative to each town), the livability score tracks relative price: rank correlation +0.21 across all 300 (95% interval +0.10 to +0.31) and +0.25 on hill land (194 localities, +0.12 to +0.38); on plains and foothills there is no clear link (+0.14, interval includes zero). Services and access carry most of it. Among the hazard drivers, the landslide driver leans towards a discount on hill land (−0.17, interval −0.31 to −0.01), and so does total hazard (−0.15, interval −0.29 to −0.01); the earthquake driver follows region, not price. The event-fitted score shows no price discount (+0.09 on hill land, interval −0.04 to +0.23), because towns with higher prices also have more recorded events. Many localities have a single listing, some places could not be located, and outside Dehradun the hill link is weaker (+0.06, interval −0.10 to +0.22; +0.18 using the largest portal alone), so these are consistency checks, not a validation.

**What did not hold up.** The national landslide-susceptibility map and the rule-based “Severe” band showed almost no skill on independent events; “safe buildable ground” (slope ≤25°) was not shown to be safer; river proximity is a *risk marker* (events cluster beside streams), not a safety signal.

## Not in this report yet

- Climate projections by scenario and year
- Ground subsidence from radar
- Land purchase, zoning and building rules (the diligence note covers what to check)
- Structural and earthquake-engineering assessment
- Electrification status and quality of supply
- Floods after 2021 in the observed-inundation record

## Known limits of the method

- The landslide rank is relative: it orders patches against similar ground, it does not give a probability or a guarantee, and within one district it separates plots only modestly.
- Flood-basin classes use fixed height-above-channel rules at 30 m; they are not hydraulic flood models and give no depth or timing.
- Observed monsoon inundation covers low-lying ground for 1984 to 2021; narrow gorges and floods that left no cloud-free satellite scene are missed, and floods after 2021 are not in it.
- Short, intense cloudburst rain is not a modelled input: satellite rainfall grids of 5 to 11 km cannot resolve a cloudburst cell.
- Glacial-lake and glacier outburst corridors are traced from mapped lakes and glaciers and cut off at each data block's edge.
- Roads, power lines and banks come from OpenStreetMap, which is patchy for minor roads and power lines in remote hills; distances are straight-line.

- Distance to the nearest hospital or school follows the road network, but depends on OpenStreetMap carrying that building at all; small-settlement government schools and clinics are tagged less consistently than private or urban ones, so a larger figure can mean the nearest one isn't mapped yet, not that none exists.
- District capital and major commercial centre are measured against a short list of large town centres per state, not every local market town, so two patches in the same district can show the same distance.
- Landslide-dam breach discharge is modelled for the West block (Uttarakhand west of the Alaknanda) only.
- Scenario flood reach and depth below a glacial lake use mapped lakes of 5,000 m² and larger (Hi-MAG 2017, Sentinel-2 2022), a volume-area law, a 30 m terrain model that cannot resolve gorges narrower than about 30 m, and flow rates not tested against measured events; read the range, not the central value alone.
- Landslide-dam scenarios cover 180 screened candidate sites (60 per block, a 50 m dam at the steepest, most landslide-prone channel reaches), not every possible dam.
- Recorded events come from news and official lists, are located to village level at best, and favour roads and towns.

## Method and sources

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Each patch is a hexagon of about 0.8 km². Terrain, slope and drainage come from the Copernicus 30 m elevation model; flood basins from height above the nearest river or torrent channel; outburst corridors are traced downstream from mapped glacial lakes and glaciers. The landslide rank is a model trained on the Geological Survey of India landslide inventory (Himachal) and tested on recorded landslides found afterwards. Observed monsoon inundation is the share of July–August Landsat observations (1984–2021) showing water, graded against independent flood maps. Every threshold lives in one rules file with its reason, and every figure in this report is stored in `figures.json` beside it.

| SOURCE | USED FOR | TERMS |
|---|---|---|
| Copernicus DEM GLO-30 | Elevation, slope, drainage, ground map | Copernicus DEM licence |
| ILSM landslide susceptibility (IIT Delhi) | Landslide share | Research dataset — see source |
| Hi-MAG glacial lakes (Chen et al. 2021) | Outburst corridors | CC BY 4.0 |
| Randolph Glacier Inventory | Glacier debris corridors | See source |
| JRC Global Surface Water (monthly history) | Historical open water; observed monsoon inundation | Copernicus / JRC terms |
| Global Flood Database (Cloud to Street) | Calibration of inundation grades | CC BY 4.0 |
| Geological Survey of India landslide inventory (Himachal subset) | Training of the landslide rank | CC BY 4.0 (Mendeley Data redistribution) |
| News and agency reports of landslides, floods and cloudbursts | Independent testing of the landslide rank | Per-article sources kept on file |

| SOURCE | USED FOR | TERMS |
|---|---|---|
| WRI Aqueduct Floods | 100-year flood depth (context) | See source |
| Sentinel-2 2022 glacial-lake inventory (Kumar and Vijay), added to Hi-MAG | Lakes formed since 2017; scenario flood depth (volume-area law, Evans 1986 peak discharge, height-above-drainage depth) | CC BY 4.0 |
| Recorded events register (news and official lists, 2010-2026) | Recorded events near the plot | Per-article sources kept on file |
| OpenStreetMap | Roads, railways, services, shops | ODbL |
| Google Open Buildings temporal | Building growth | CC BY 4.0 |
| WorldPop · VIIRS night lights · ERA5-Land · MODIS | Population, activity, climate context | See each source |
| Documented incidents compilation | Incident flags | GSI, NIDM, NDMA, USDMA records |